C1. The auction buyer

Your client. Viktor buys about 40 cars a month at auction and resells them.

Your deliverable. A recommended maximum bid for any listing, plus a clear statement of how conservative that number is and why.

How to proceed

1. Clean the data, fit a price model on `log(price_eur)`, check it beats your baseline.
2. Work out, from the two costs alone, which percentile of the predicted distribution a rational bidder should target. There is a short derivation. Do it before you write any code for step 3.
3. Now measure it. Sweep several candidate percentiles, and for each one compute what Viktor actually loses on the test set: mean euros overpaid and mean euros of margin lost. Combine them with his 900 and 150. The cheapest one is your answer.
4. Compare the two. If your derivation and your sweep agree, say so. That agreement is worth more than either number alone.
5. Convert everything back to euros before costing it. Costing in log space is the single most common error in this brief.

If you have not met quantile regression

You do not need it. Fit an ordinary regression on `log(price_eur)`, hold out a slice of your **training** data that the model never sees, and look at the errors it makes there. The percentiles of those errors are your offsets: add the 10th percentile error to a prediction and you have a lower bound, add the 90th and you have an upper bound. This is Day 2 material and it gets essentially the same answer as `sklearn.linear_model.QuantileRegressor`. Either route is accepted. If you use `QuantileRegressor`, set `alpha=1e-3`; the default flattens the fit.

What earns what

- **Pass.** A measured sweep, the cheapest percentile chosen, the bid reported in euros.
- **Good.** Plus the derivation done first and confirmed by the sweep, plus mean overpay and mean shortfall stated separately so Viktor can see both failure modes.
- **Excellent.** Plus sensitivity: Viktor's 900 is an estimate, not a measurement. Show how your recommended bid moves if it is really 600, or 1,500. Expect to be asked this.

What fails. Fitting an ordinary regression and subtracting a safety margin that felt about right. The first question at the defence will be where the margin came from.

Required of every team

These are 45 of the 100 marks. They are the same for all seven briefs, and they are where most teams lose points, not on the modelling.

1. **A baseline** (10). The simplest thing that could serve your client, scored on the same data. Your system must beat it, and you must say by how much. "We didn't build one" is zero.
2. **An honest evaluation** (15). Split the data before you look at it. Score the test set once, at the end. If you tried five models and picked the best on the test set, that number is not real. Never use `days_on_market` as an input: it is only known after the car sells.
3. **A number in your client's units** (10). Euros, cars, percent of listings. Not R squared.
4. **A trust check** (10). Your client will send inputs unlike anything you trained on. Show that your system detects this, and say what it does about it. Check that the rows you flag really are worse; if they are not, your detector is decoration.
5. **Known failures** (10). At least two specific segments where you are systematically wrong, with the direction, the size and the reason. "The model is not perfect" earns nothing.

The remaining 55: your client's decision (25), the defence (15, both members answer), your audit question to the team you check (5). Full criteria are in the rubric handout.